Help for command VisuPose3D

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1 Descrition of VisuPose3D

This command create a $3d$ visualisation of poses and also, possibly, a sparse visualisation of the scene (using tie points). The visualisation is made using point cloud in ply format because it can be read very easily on various free open source software.

The poses of camera are visualized using a pyramid for rectilinear camera and set of "meridian" for fisheye or lidar station (see 1 , a picture is worth a thousand words).

2 fixing size of pyramid

The size of the pyramid is proportionnal to the $35mm$ focal length (precisely the focal in pixel divided by diagonal of image).

The default rule to compute this degree of proportionnality uses the parameter `ELPFD` [1] and the tie points :

- a depth D is computed using a proportion of all the depth of all tie points (default 30%);
- this depth D is mutilpied by a factor (default $\frac{1}{10}$).

If the value of parameter `CamScale` is set it is used instead of the previous computation. The default value of `CamScale` ($\frac{1}{10}$) is also used if there is no tie-points .

[1] Evaluate Lenght Pyramid From Depth ...

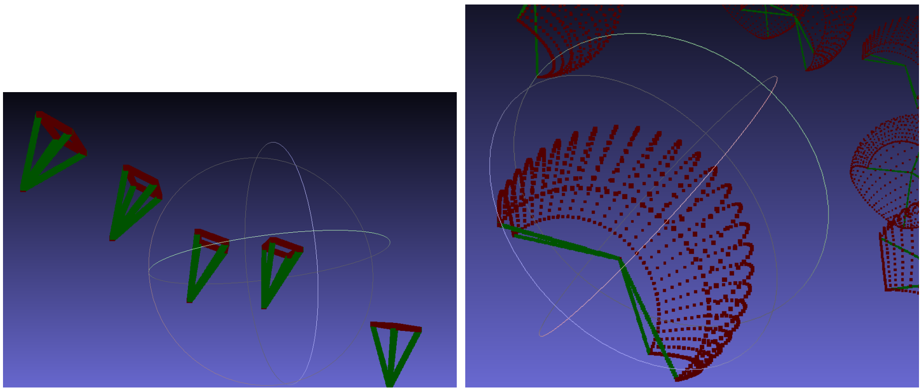

Figure 1: Result of VisuPose3Dwith "standard" camera and fisheye